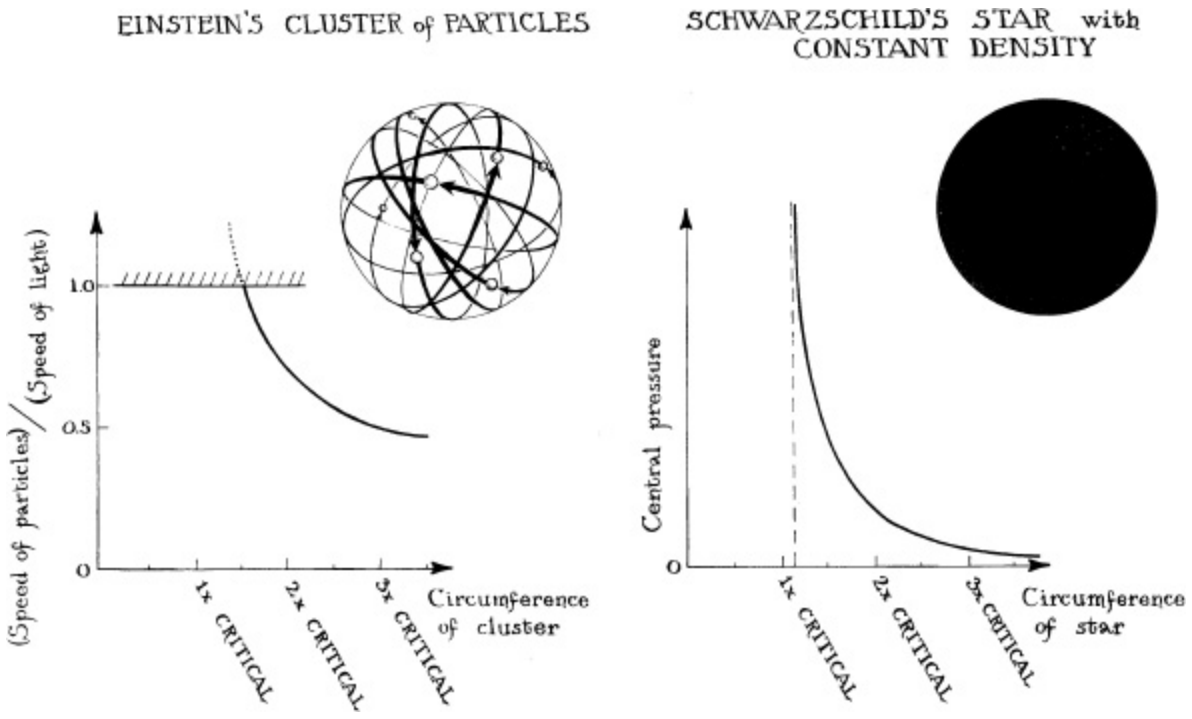
particles that pull on each other gravitationally and thereby hold the cluster together, in much the same way as the Sun holds the solar system together by pulling gravitationally on its planets. The particles in Einstein's cluster all moved in circular orbits around a common center; their orbits formed a sphere, with particles on one side of the sphere pulling gravitationally on those on the other side (left half of Figure 3.5).



3.5 Einstein's evidence that no object can ever be as small as its critical circumference. *Left:* If Einstein's spherical cluster of particles is smaller than 1.5 critical circumferences, then the particles' speeds must exceed the speed of light, which is impossible. *Right:* If a star with constant density is smaller than 9/8 = 1.125 critical circumferences, then the pressure at the star's center must be infinite, which is impossible.

Einstein imagined making this cluster smaller and smaller, trying to drive its actual circumference down toward the critical circumference. As one might expect, his calculation showed that the more compact the cluster, the stronger the gravity at its spherical surface and the faster the particles must move on its surface to prevent themselves from being pulled in. If the cluster were smaller than 1.5 times the critical circumference, Einstein's calculations showed, then its gravity would be so strong that the particles would have to move faster than the speed of light to avoid being pulled in. Since nothing can move faster than light, there was no way the cluster could ever be smaller than 1.5 times critical. "The essential result of this